INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Selected topics in Cryptography

Quiz 3A *March 12, 2026*

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Calif 3.5/5

Problems

Read the following problem and solve it. Please write down the detail of all your calculations .

1. Alice and Bob are running the ECDH, they are using the elliptic curve $E : y^2 = x^3 + 3x + 2$ mód 11. The points in this curve are $E = \{\mathcal{O}, (2, \pm 4), (3, \pm 4), (4 \pm 1), (6, \pm 4), (7, \pm 5), (10, \pm 3)\}$. Answer the following questions: $\#E = 13$

   0.5 a) Which point is a generator G in this curve? Why? (**Correct answer: 1 point**) (2, 4) why?

   3/4 b) Run the ECDH protocol for Alice and Bob. Use the generator G that you found in the previous point. Suposse that Alice have chosen the secret $a = 2$ and Bob $b = 3$. Compute the intermediate values and the secret K agreed between Alice and Bob. (**Correct answer: 4 points**) a = 3, b = 2, p = 11, G = (2, 4)

   k.a = 2 k.b = 3

   2G = (10, 3)
   3G = (4, 10)

Useful mathematics

> **Elliptic Curve Point Addition and Point Doubling**
>
> $$x_3 = s^2 - x_1 - x_2 \bmod p$$
> $$y_3 = s(x_1 - x_3) - y_1 \bmod p$$
>
> where
>
> $$s = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} \bmod p \text{ ; if } P \neq Q \text{ (point addition)} \\ \frac{3x_1^2 + a}{2y_1} \bmod p \text{ ; if } P = Q \text{ (point doubling)} \end{cases}$$

2(4, 9)
S = (3·16 + 3)(2·9)$^{-1}$
= (51)(18)$^{-1}$ mod 11
=

x_1 y_1
2(2, 4) = (10, 8) = (10, -3)
S = (3·4 + 3)(2·4)$^{-1}$ mod 11
= (15)(8)$^{-1}$ mod 11
= 4(7) mod 11
= 28 mod 11 = 6
x_3 = 36 − 2 − 2 = 32 mod 11
= 10
y_3 = 6(2 − 10) − 4 mod 11
= −52 mod 11
= 11 − 52 mod 11 = 11 − 8 = 3

Error aritmético

3(2, 4) = (10, 3) + (2, 4)
S = $\frac{4-3}{2-10}$ = 1(−8)$^{-1}$ mod 11
= (11 − 8 mod 11)$^{-1}$ = (3)$^{-1}$ mod 11
= 4
x_3 = 16 − 10 − 2 = 4
y_3 = 4(10 − 4) − 4 = 4(6) − 4
= 24 − 4 = 20 mod 11 = 9

Alice
k.a = 2
G = (2, 4)
A = 2G = 2(2, 4) = (10, 3)

k = 2(4, 9)

y K?

Bob
k.b = 3
G = (2, 4)
B = 3G = 2P + P = (10, −3) + (2, 4)
= (4, 9)

k = 3(10, 3)

error aritmético
3G = (4, 10)